Roots of Polynomials 1

Core Pure Mathematics 1

Chapter 4

Roots of Polynomials

Chapter Overview 2

This chapter relates polynomial coefficients, such as a, b, c in $ax^2 + bx + c$, to its roots: values where the polynomial is zero.

1. Relationships for quadratics, cubics and quartics

Given $kx^2 + (k-3)x - 2 = 0$, find k when the roots sum to 4.

2. Expressions in the roots

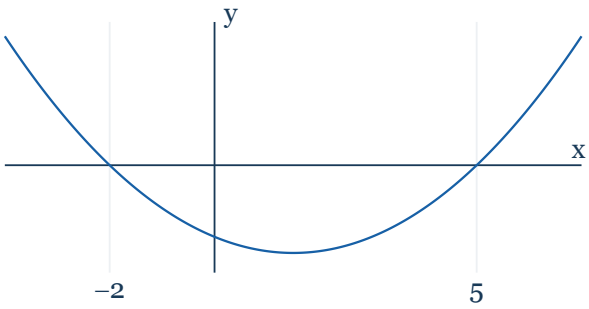
Without finding the roots of $x^3 - 3x^2 - 3x + 1 = 0$, find the sum of their squares.

3. Linear transformations of roots

If $x^4 - 3x^3 + 15x + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$, find an equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1, 2\delta + 1$.

Introduction 3

$p(x) = x^2 - 3x - 10$ has roots $-2, 5$.



We want to understand how roots relate to coefficients.

What do you notice about the sum $-2 + 5 = 3$?

It is the negative of the coefficient of x.

What do you notice about the product $(-2)(5) = -10$?

It equals the constant term.

At GCSE, factorising a monic quadratic used two numbers whose sum gives the middle coefficient and product the constant. The roots are the negatives of those factor constants.

Introduction 4

Let α, β be roots of $ax^2 + bx + c = 0$, $a \neq 0$.

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$
$$= a[x^2 - (\alpha + \beta)x + \alpha\beta]$$
$$= ax^2 - a(\alpha + \beta)x + a\alpha\beta.$$

Compare coefficients:

$$b = -a(\alpha + \beta), \qquad c = a\alpha\beta.$$

Quadratic roots

$$\alpha + \beta = -b/a, \qquad \alpha\beta = c/a.$$

Higher-degree Polynomials 5

There are similar relationships for higher degrees; always assume the leading coefficient is nonzero.

Quadratic: $ax^2 + bx + c$

Roots α, β.

$\alpha + \beta = -b/a, \quad \alpha\beta = c/a.$

Cubic: $ax^3 + bx^2 + cx + d$

$\alpha + \beta + \gamma = -b/a$

$\alpha\beta + \beta\gamma + \gamma\alpha = c/a$

$\alpha\beta\gamma = -d/a$

Quartic: $ax^4 + bx^3 + cx^2 + dx + e$

$\alpha + \beta + \gamma + \delta = -b/a$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = c/a$

$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -d/a$

Spot the pattern. What is the product of all four roots?

$\alpha\beta\gamma\delta = e/a.$

For a quintic the product is $-f/a$; for a sextic it is g/a. Signs alternate. These are Vieta's formulae.

Back to roots of quadratics... 6

The roots of $2x^2 - 5x - 4 = 0$ are α, β. Without solving, find (a) their sum, (b) their product, (c) $1/\alpha + 1/\beta$, (d) $\alpha^2 + \beta^2$.

Use the sum and product to calculate these expressions without needing the individual roots. For the last two, rewrite in terms of $\alpha + \beta$ and $\alpha\beta$.

a

b

c

d

Further Example 7

The quadratic $ax^2 + bx + c = 0$ has roots $-3/2, 5/4$. Find integer a, b, c.

Initially choose $a = 1$; rescale to make all coefficients integers.

1

Test Your Understanding 8

For $x^2 + 2x + 3 = 0$, find (a) the root sum, (b) the root product.

1a

1b

A quadratic has roots $2/3, 1/5$. Find integer coefficients a, b, c.

2

8. $kx^2 + (k-3)x - 2 = 0$. Find k if its roots sum to 4.

8

9. $nx^2 - (16+n)x + 256 = 0$ has real roots $\alpha, -\alpha$. Find n.

9

10. The roots of $6x^2 + 36x + k = 0$ are reciprocals. Find k.

10

11. $mx^2 + 4x + 4m = 0$ has roots $k, 2k$. Find m, k.

11

12. $ax^2 + 8x + c = 0$, with real a, c, has roots α, α^*.

a. Given $\text{Re}\,\alpha = 2$, find a.

b. Given $\text{Im}\,\alpha = 3$, find c.

a

b

13. $4x^2 + px + q = 0$, with real p, q, has roots α, α^*.

a. Given $\text{Re}\,\alpha = -3$, find p.

b. If $\text{Im}\,\alpha \neq 0$, find the range of q.

a

b

Roots of Cubics 13

A cubic $ax^3+bx^2+cx+d=0$, $a\neq 0$, has three roots counting multiplicity. For real coefficients, either all roots are real or one is real and two are non-real conjugates.

$$\begin{aligned}&a(x-\alpha)(x-\beta)(x-\gamma)\\&=ax^3-a\alpha x^2-a\beta x^2-a\gamma x^2\\&\quad+a\alpha\beta x+a\beta\gamma x+a\gamma\alpha x-a\alpha\beta\gamma\\&=ax^3-a(\alpha+\beta+\gamma)x^2\\&\quad+a(\alpha\beta+\beta\gamma+\gamma\alpha)x-a\alpha\beta\gamma.\end{aligned}$$

Comparing coefficients:

$$\alpha+\beta+\gamma=-b/a$$
$$\alpha\beta+\beta\gamma+\gamma\alpha=c/a$$
$$\alpha\beta\gamma=-d/a.$$

The sum rule matches the quadratic rule. For quadratics there is only one pair product. For cubics there is only one triple product, the product of all three roots.

Examples 14

The roots of $2x^3+3x^2-4x+2=0$ are α,β,γ. Without solving, find (a) their sum, (b) their pair-product sum, (c) their product, (d) the sum of their reciprocals.

For the reciprocals, rewrite using the preceding symmetric expressions.

a

b

c

d

Examples 15

A cubic has roots $1-2i, 1+2i, 2$. Find integer coefficients a, b, c, d.

As before, first choose a leading coefficient of 1.

1

Exercise 4B - Core Pure Book 1 Pages 58-59 16

8. $2x^3 - kx^2 + 30x - 13 = 0$ has roots α, β, γ.

a. State the pair-product sum and product, and express k in terms of the roots.

b. Given $\alpha = 2 - 3i$, find k.

a

b

9. $x^3 - mx + n = 0$ has roots $1, -4, \alpha$.

a. State, with a reason, whether α is real.

b. Find m, n, α.

a

b

10. $2x^3 - 10x^2 + 8x - k = 0$ has root $3 - i$.

a. Find the other roots.

b. Find k.

a

b

11. $x^3 - 14x^2 + 56x - 64 = 0$ has roots $\alpha, k\alpha, k^2\alpha$, with real k. Find α, k.

1

2

12. $8x^3 + 12x^2 - cx + d = 0$ has roots $\alpha, \alpha/2, \alpha - 4$. Find α, c, d.

12

Challenge: Why the Expressions Are Real 21

α, β, γ are roots of a real-coefficient cubic. Considering all real/non-real possibilities, explain why (a) their sum, (b) their pair-product sum, (c) their product are real.

Start

a

b

c

Roots of Quartics 22

Quadratic: $ax^2 + bx + c$

Roots α, β.

$\alpha + \beta = -b/a, \quad \alpha\beta = c/a.$

Cubic: $ax^3 + bx^2 + cx + d$

$\alpha + \beta + \gamma = -b/a$

$\alpha\beta + \beta\gamma + \gamma\alpha = c/a$

$\alpha\beta\gamma = -d/a$

Quartic: $ax^4 + bx^3 + cx^2 + dx + e$

$\alpha + \beta + \gamma + \delta = -b/a$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = c/a$

$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -d/a$

For the quartic, the product of all four roots is $\alpha\beta\gamma\delta = e/a$.

Count each distinct pair and triple exactly once.

Example 23

$x^4 + 2x^3 + px^2 + qx - 60 = 0, p, q \in \mathbb{R}$, has roots $\alpha, \beta, \gamma, \delta$, with $\gamma = -2 + 4i, \delta = \gamma^*$.

a. Show $\alpha + \beta - 2 = 0, \alpha\beta + 3 = 0$.

b. Find all roots and p, q.

$\mathbb{C}$ denotes complex numbers. A notation such as $\Sigma\alpha\beta$ means the sum over all distinct pairs.

a

b

Exercise 4C - Core Pure Book 1 Pages 60-61 24

8. $x^4 - 16x^3 + 86x^2 - 176x + 105 = 0$ has roots $\alpha, \alpha + k, \alpha + 2k, \alpha + 3k$, with real k. Solve the equation.

1

2

9. $3072x^4 - 2880x^3 + 840x^2 - 90x + 3 = 0$ has roots $\alpha, r\alpha, r^2\alpha, r^3\alpha$, with real r. Solve the equation.

1. Pair products

2. Triple products

3. Solve

10. Three roots of $40x^4 + 90x^3 - 115x^2 + mx + n = 0$ are $1, -3, 1/2$.

a. Find the fourth root.

b. Find m, n.

a

b

11. $2x^4 - 34x^3 + 202x^2 + dx + e = 0$ has roots $\alpha, \alpha + 1, 2\alpha + 1, 3\alpha + 1$.

a. Find α.

b. Find d, e.

a

b

12. $4x^4 - 19x^3 + px^2 + qx + 10 = 0$, with real p, q, has roots $\alpha, \beta, 3 + i, 3 - i$.

a. Show $4\alpha + 4\beta + 5 = 0, 4\alpha\beta - 1 = 0$.

b. Find all roots and p, q.

c. Plot the roots on an Argand diagram.

a

b(ii)

b(i)

c

Expressions in the Roots 29

We can calculate symmetric expressions without finding individual roots. For a quadratic:

$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta,$

$\dfrac{1}{\alpha}+\dfrac{1}{\beta}=\dfrac{\alpha+\beta}{\alpha\beta}\quad(\alpha\beta\neq 0).$

Sums of squares

For any number of roots, square the root sum and subtract twice the sum of distinct pair products.

Cubic:

$\alpha^2+\beta^2+\gamma^2=S_1^2-2S_2.$

Quartic:

$\alpha^2+\beta^2+\gamma^2+\delta^2=S_1^2-2S_2.$

Sums of cubes

Quadratic:

$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta).$

Cubic:

$\alpha^3+\beta^3+\gamma^3=S_1^3-3S_1S_2+3S_3.$

Here S_j sums distinct products of j roots. These results may be used without proof.

The same symmetric-sum pattern extends to higher degrees. A quartic sum-of-cubes formula is not required in this chapter.

Example 30

Three roots satisfy $\alpha\beta\gamma=4$, $\alpha\beta+\beta\gamma+\gamma\alpha=-5$, $\alpha+\beta+\gamma=3$. Find $(\alpha+3)(\beta+3)(\gamma+3)$.

Expand and group into the known symmetric expressions.

1

11. The roots of $2x^3 + 4x^2 + 7 = 0$ are α, β, γ.

a. State their sum, pair-product sum and product.

b. Find (i) $\alpha^2 + \beta^2 + \gamma^2$, (ii) $\alpha^3\beta^3\gamma^3$, (iii) $(\alpha + 2)(\beta + 2)(\gamma + 2)$.

a

b(i)

b(ii)

b(iii)

12. Prove

$$\alpha^3 + \beta^3 + \gamma^3 = S_1^3 - 3S_1S_2 + 3S_3,$$

where $S_1 = \alpha + \beta + \gamma$, $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha$, $S_3 = \alpha\beta\gamma$.

1

2

13. $3x^3 - px + 11 = 0$ has roots α, β, γ.

a. If the pair-product sum is 4, find p.

b. State the root sum and product.

c. Find $(3-\alpha)(3-\beta)(3-\gamma)$.

a

b

c

14. $x^4 + 2x^2 - x + 3 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a. State S_1, S_2, S_3, S_4.

b. Find (i) the sum of reciprocals, (ii) the sum of squares, (iii) $(\alpha+1)(\beta+1)(\gamma+1)(\delta+1)$.

a

b(i)

b(ii)

b(iii)

15. $ax^4 + 3x^3 + 2x^2 + x - 6 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a. If their product is -3, find a.

b. State S_1, S_2, S_3.

c. Find the sum of reciprocals.

a

b

c

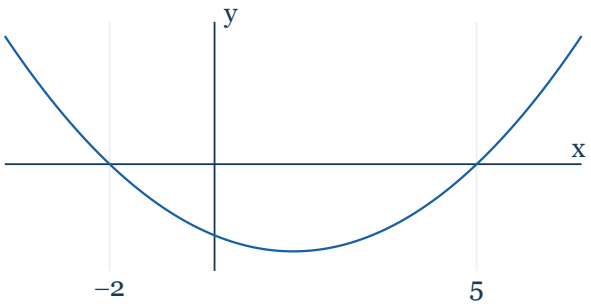
16. Prove that for four roots,
$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = S_1^2 - 2S_2$.

16

Linear Transformations of Roots 37

Original polynomial:

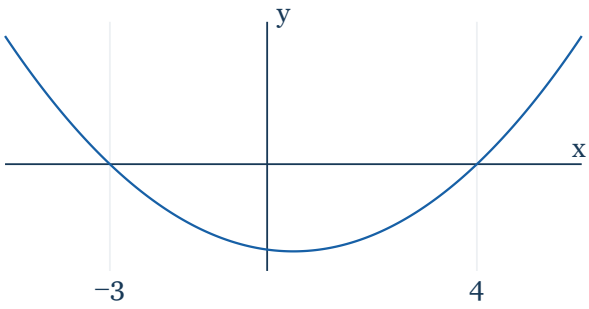
$p(x) = (x+2)(x-5) = x^2 - 3x - 10.$



How do the coefficients change when each root decreases by 1?

Transform each root by $\alpha \mapsto \alpha - 1$:

$(x+3)(x-4) = x^2 - x - 12.$



In factorised form, replace each root by one less. The effect on expanded coefficients is less immediate.

Continuing this example 38

Without finding the roots α, β of $x^2 - 3x - 10 = 0$, find an equation with roots $\alpha - 1, \beta - 1$.

Method 1: coefficients

Method 1

Method 2: substitution

Use a new variable for the transformed root.

Method 2

This is the familiar function transformation: translating roots left by 1 means using $f(w+1)$. Substitution is generally the simpler method.

Quartic Example 39

The roots of $x^4 - 3x^3 + 15x + 1 = 0$ are $\alpha, \beta, \gamma, \delta$. Find an equation whose roots are twice each original root plus 1.

Method 1: symmetric sums

Method 1

Method 2: substitution (recommended)

Method 2

Test Your Understanding 40

$x^3 - 2x^2 + 4 = 0$ has roots α, β, γ. Find an equation with roots $3\alpha - 1, 3\beta - 1, 3\gamma - 1$.

Method 1: symmetric sums

Method 1

Method 2: substitution

Method 2

3. Without solving $x^3 - 3x^2 + 4x - 7 = 0$, find an equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1$, in the form $aw^3 + bw^2 + cw + d = 0$ with integer coefficients.

3

4. Without solving $x^3 + 4x^2 - 4x + 2 = 0$, find an equation with roots $2\alpha - 1, 2\beta - 1, 2\gamma - 1$, in the form $w^3 + pw^2 + qw + r = 0$ with integers p, q, r.

4

5. Without solving $3x^3 - x^2 + 2x - 5 = 0$, find an equation with roots $3\alpha + 1, 3\beta + 1, 3\gamma + 1$, with integer coefficients.

5

6. $2x^4 + 4x^3 - 5x^2 + 2x - 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find integer-coefficient equations whose roots are (a) three times the original roots, (b) each original root minus 1.

a

b

7. Without solving $x^4 + 2x^3 - 3x^2 + 4x + 5 = 0$, find integer-coefficient equations with roots (a) twice each original root, (b) each original root minus 2.

a

b

8. Without solving $3x^4 + 5x^3 - 4x^2 - 3x + 1 = 0$, find integer-coefficient equations with roots (a) three times each original root, (b) each original root plus 1.

a

b

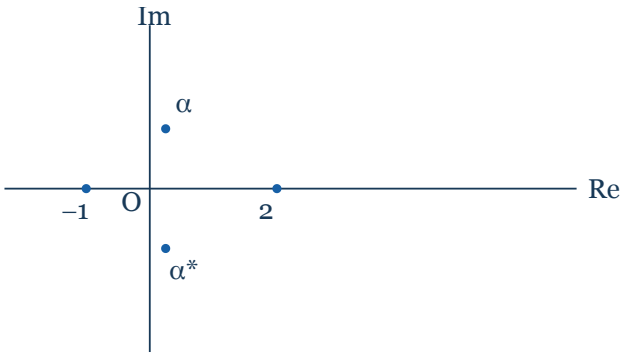
Extension 47

The quartic $2x^4 - 3x^3 + x^2 - 2x - 6 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a. Find an integer-coefficient equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1, 2\delta + 1$.

a

b. The diagram shows the original roots. Sketch the roots after the transformation.



b

Why These Relationships Are Useful 48

What do sums and products tell us?

They connect the geometry of roots with polynomial coefficients. The mean of n roots is S_1/n; on an Argand diagram this is the centroid of their points, counting multiplicity.

For $z^k = a + ib$, integer $k \geq 2$, the sum of the roots is zero because the coefficient of z^{k-1} is zero.

Why is this useful?

If one root is missing, subtract the known roots from $-b/a$. If a coefficient is unknown, use information about the roots to determine it. These relationships strengthen our ability to reason about polynomials.

Why sums of squares or reciprocals?

They let us evaluate expressions without solving for individual roots, and form equations for transformed roots. This algebraic structure remains useful even when no immediate physical application is given.